

Roll No.: _____

Pandit Deendayal Energy University**School of Technology**

End Semester Examination

B. Tech. (Computer Science and Engineering)

Semester –VI**Course Name :** Cryptography and Network Security**Course Code :** 23CP314T**Date:** 10/05/2024**Time:** 3 hours**Max. Marks:** 100 marks**Instructions:**

1. Do not write anything other than your roll number on question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

	Max. Marks (10)	CO Attained CO5
1. For the following program segment the webpage provides output based on the user input provided. Which type of attack is possible on this program segment? Give example and also discuss the coding technique to withstand against that attack.		
<pre> uName = getRequestString("username"); uPass = getRequestString("userpassword"); sql = 'SELECT * FROM Users WHERE Name =' + uName + ' AND Pass =' + uPass + ' ' </pre>		
2. Two well-known public key encryption algorithms are there: RSA and El-Gamal. Discuss which one is deterministic encryption algorithm and which one is randomized encryption algorithm. Justify your decision.	(10)	CO2
3. Let the faculty has to convey the grade for a student in a confidential manner using the public key of student in RSA setup. The grade point is a value between 0 to 10.	(10)	CO4
• Which type of attack is possible on this system? Explain with reasons. • Whether the same attack is possible with El-Gamal cryptosystem? Why?		
4. Compute $A(x) + B(x) \bmod P(x)$ and $A(x) * B(x) \bmod P(x)$ in $GF(2^4)$. $P(x) = x^4 + x + 1$	(10)	CO3
• $A(x) = x^2 + 1$, $B(x) = x^3 + x^2 + 1$ • $A(x) = x^2 + 1$, $B(x) = x + 1$		
5. Using the extended Euclidean algorithm, compute the greatest common divisor and the parameters s, t of	(10)	CO3
• 198 and 243 • 1819 and 3587		

OR

5. Compute the following exponentiations $x^e \bmod m$ applying the square-and-multiply (Fast exponentiation) algorithm: (10) CO3
 $x = 2, e = 79, m = 101$
 $x = 3, e = 197, m = 101$
6. The SSL Protocol aids security in http Protocol. Justify this statement. (10) CO1
Describe the handshake in SSL Protocol.
- OR
6. For a secure communication over public channel, we need to use the combination of public and symmetric key encryption. Is it true? Why? (10) CO1
7. List the two tests that a pseudorandom number generator must have to pass to become a cryptographically secure random number generator. (10) CO6
8. **Do any two** from the following three. (10) CO3
- a) List all the points generated by an elliptic curve group defined with the equation $E: y^2 = x^3 + 5 \pmod{7}$.
- b) In a (3,5) Shamir secret sharing scheme with modulus $p = 17$, the following were given to Alice, Bob, and Charles: (1,8), (3,10), (5,11). Calculate the equation and identify the secret
- c) Use the Chinese Remainder Theorem to find an x such that
 $x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}, x \equiv 10 \pmod{11}$
9. In a (3,5) Shamir secret sharing scheme with modulus $p = 17$, the following were given to Alice, Bob, and Charles: (1,8), (3,10), (5,11). Calculate the corresponding Lagrange interpolating polynomial, and identify the secret. (10) CO3
10. Compare the two scenarios: One in which you will select Identity based encryption and the other where you will use attribute based encryption. Provide the reasons for your selection in each scenario. (10) CO6